



FINE JEWELLERY · EST. MMXXVI

# OPALÉRA

*Named for the opal — the one stone whose fire holds the colour of every gem: the blue of sapphire, the blush of rose gold, the light of diamond and the lustre of pearl. Eighty handcrafted pieces, one house — and a studio to make yours truly yours.*



IFM2B10 · GROUP PROJECT · FINAL PRESENTATION · SEPTEMBER 2026

## OPALÉRA — The House of Every Gem

A full-stack B2C fine-jewellery store · Quantum Coders · 80 pieces · 92 pages · Node.js backend · manager console · live at [opalera.onrender.com](https://opalera.onrender.com)

# The Team



Tsongorera RJN

225059699

Leader · Backend developer



Thedzisani KOP

224098795

Frontend developer



Madhavan T

225139703

Backend developer ·  
Database creator



Mtshare B

225226751

Frontend developer

*Dataset: the 80-piece Kaggle jewellery database · necklaces, earrings & pendants · R 959 – R 6 799 · Source: [github.com/Thriyi17/Opalera](https://github.com/Thriyi17/Opalera)*

# Why OPALÉRA Exists

*Buying jewellery online is a gamble — fake stones, no recourse, real pieces priced beyond a normal salary. OPALÉRA answers with a maison you can trust.*



## Certified authentic

Every piece carries its own certificate of authenticity; the maison refunds in full if a stone is ever misrepresented.



## Honest prices

Direct-from-atelier pricing from R 959, VAT included, with the comparable boutique estimate and instalment framing on every piece.



## The rating system

Star ratings, written patron reviews with a verified-purchase badge, and thumbs votes — visible before you pay.

# What We Built

80

PIECES IN THE  
DATABASE

92

HTML PAGES

24

API ENDPOINTS

2

USER ROLES

64

AUTOMATED CHECKS

0

NPM DEPENDENCIES



## Node.js server

server.js — routes, script password hashing, httpOnly cookie sessions, server-side totals and stock rules, static hosting of the site



## JSON database

data/db.json — users, sessions, products, orders, reviews, wishlists, carts, resets; seeded on first run; DATA\_DIR for a persistent disk



## Shared JS layer

opalera-data.js — catalogue helpers, the API client, the live catalogue mirror, order tracking; one script for every page



## Deployed

GitHub → Render: every push to main redeploys  
opalera.onrender.com — the same server, site and API

# Every Feature Stands Complete



## Website design

one gold-on-midnight system, 92 pages



## Product catalog

service-fed, search · price bands · sort · tabs



## Database

8 collections, 80 seeded products



## Web service

24 JSON endpoints, cookie sessions



## Product management

add · edit · retire · delete, live on the storefront



## Shopping cart

quantities, stock caps, saved per account



## Transactions

promo codes, 15% VAT, stock rule, 3 pay methods



## Invoices

tax invoice per order, QR code, printable



## User management

register · login · reset · patron & manager roles

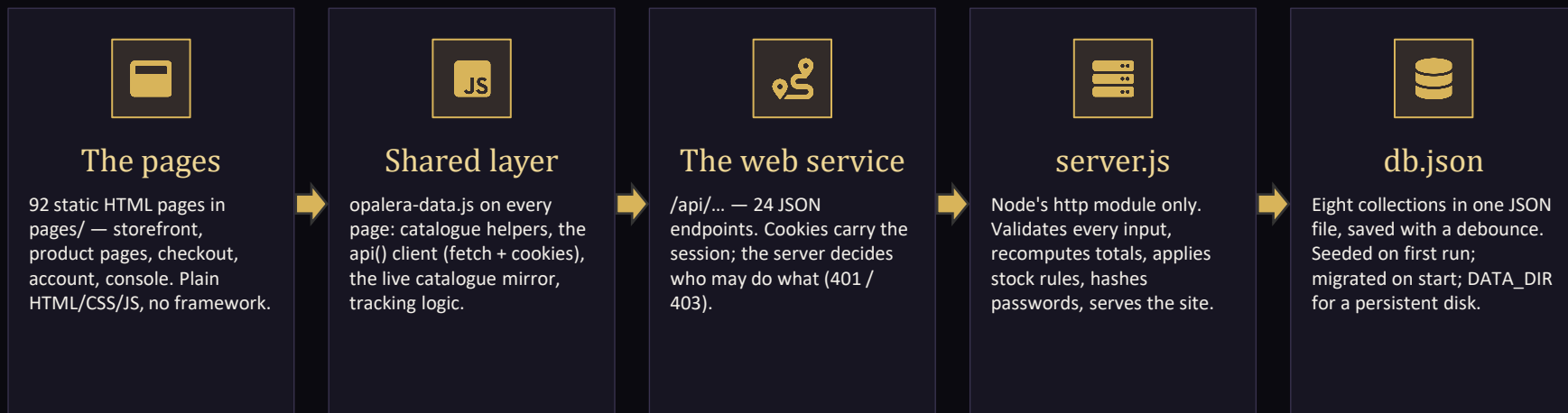


## Reports

filters, distinct-products control, stock, users/day, 6 KPIs

BONUS · compare pieces side by side · live order-tracking timeline · QR code on every invoice · the Studio customiser

# One Request, End to End



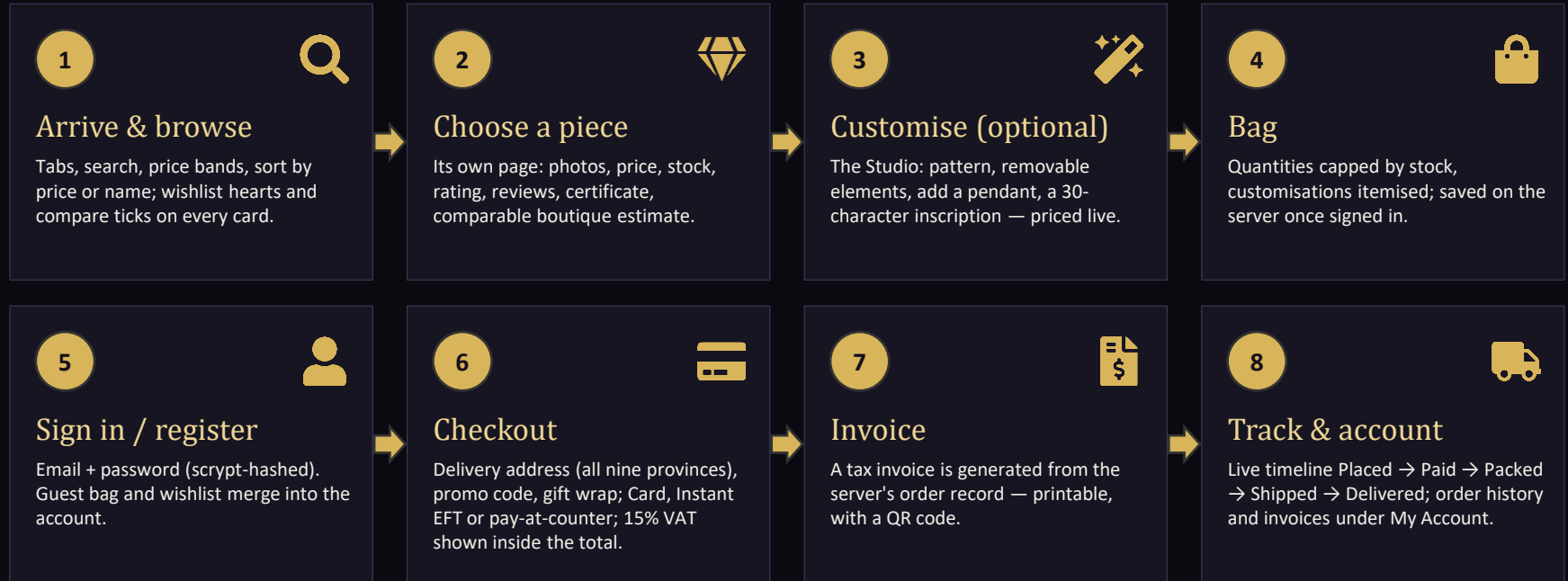
**A patron opens a product page** the page paints from the bundled catalogue instantly, then GET `/api/products` patches live price, stock and retired flags in place

**A patron checks out** POST `/api/orders` — the server recomputes item totals, discount and VAT, refuses over-ordering, deducts stock, clears the cart, stores the order

**A manager edits a piece** PUT `/api/products/:id` (manager cookie) — saved to `db.json`; every storefront tab and product page shows the change on its next fetch

**Deployment** git push to main → Render rebuilds and restarts the same server; the database re-seeds from the repo's copy

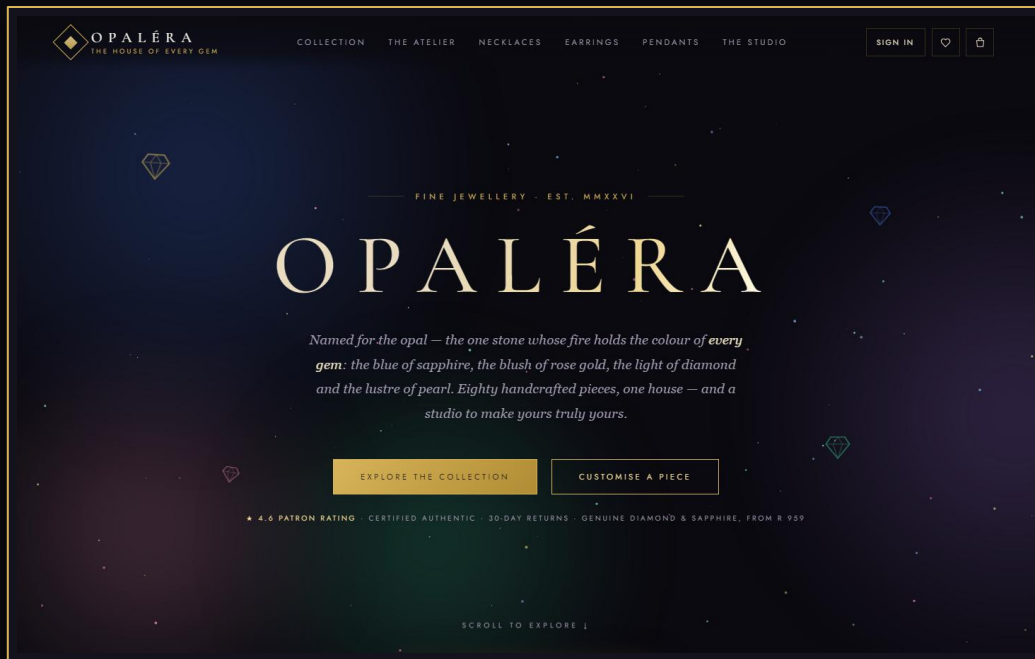
# From Arrival to Delivered, in Eight Steps



# Arrival at the Maison

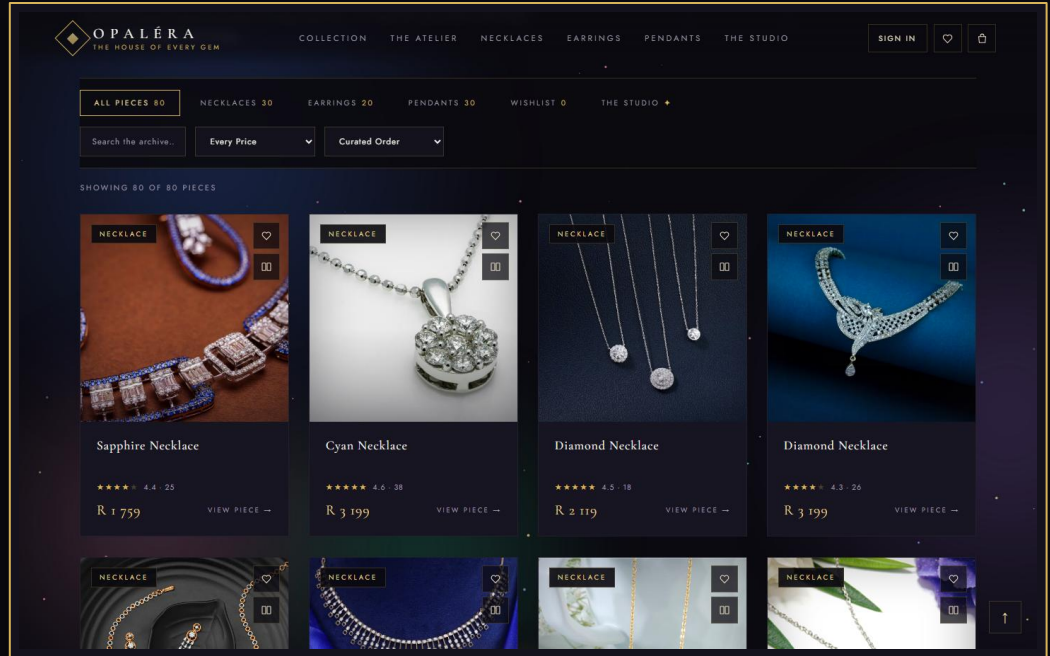
- ◆ Animated aurora, sparkle field and floating gems behind every page
- ◆ One gold-on-midnight design system across all 92 pages
- ◆ Trust strip on arrival: rating · authenticity · returns · from R 959
- ◆ Sign-in, wishlist and bag in the nav — the counts go live the moment a patron signs in

DESIGN · DONE

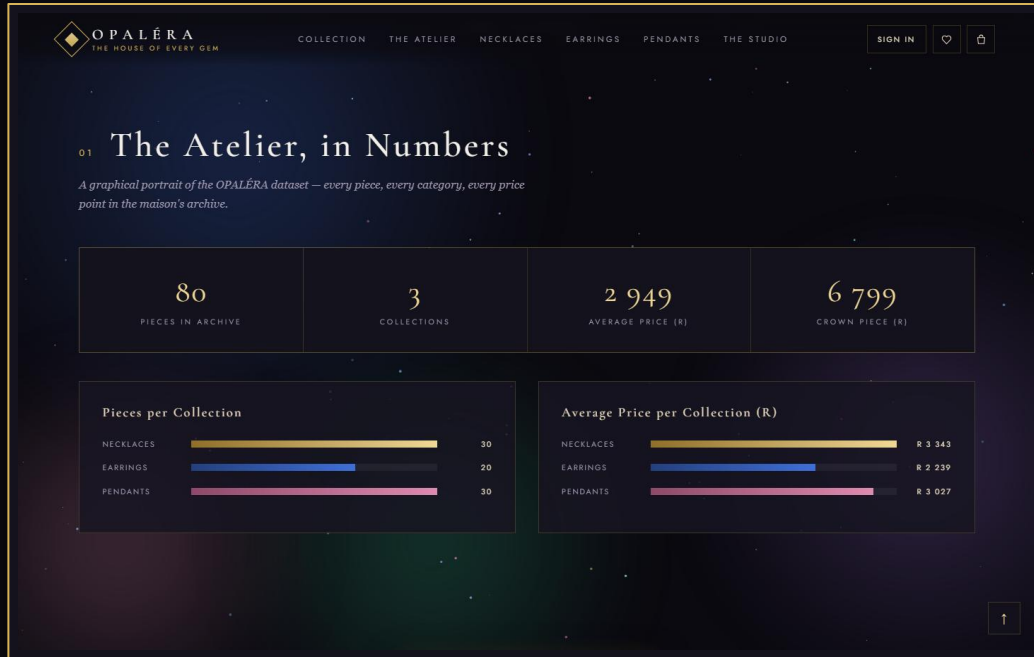


# The Collection — Fed by the Service

- ◆ The catalogue is retrieved from the database through GET /api/products — prices, photos, stock and retired pieces update live
- ◆ Category tabs with live counts, search, four price bands and sorting by price or name (the two sort criteria)
- ◆ “Showing 80 of 80 pieces” honesty line; wishlist hearts and a compare tick on every card
- ◆ Each card opens the piece's own page



# The Atelier, in Numbers



- ◆ Live stat counters computed from the catalogue: 80 pieces, 3 collections, average price R 2 949, crown piece R 6 799
- ◆ Pieces per collection and average price per collection, coloured by gem accent
- ◆ Counts animate in as the visitor scrolls
- ◆ Everything here recomputes if the manager adds or retires pieces

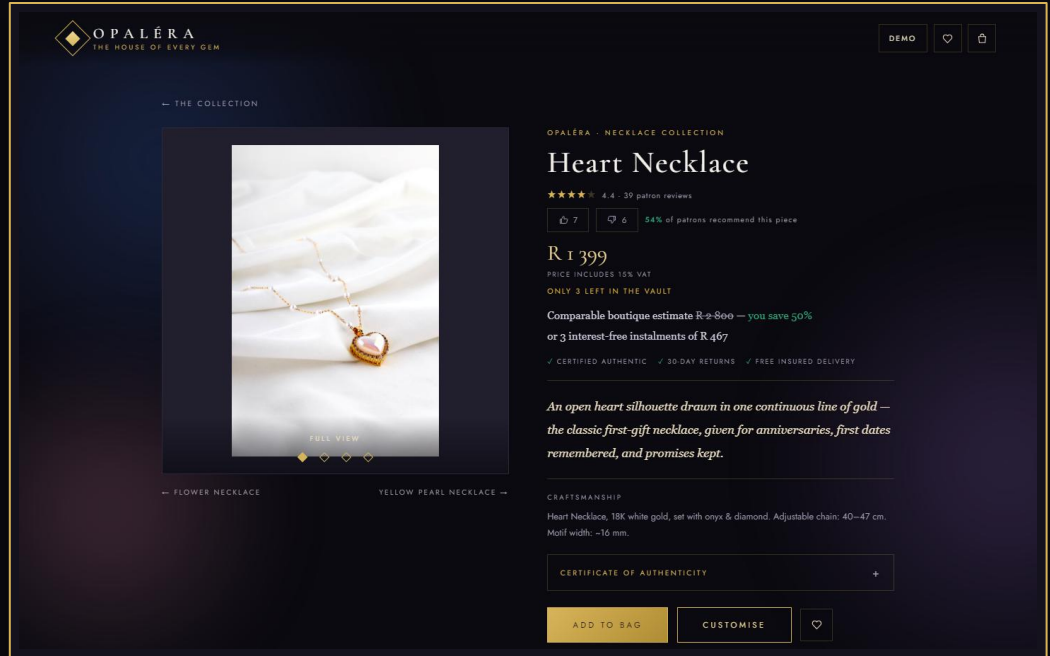
DESIGN · DONE

# The Product Page

- ◆ Hand-written description for all 80 pieces, plus craftsmanship specs
- ◆ Live stock from the database — “Only 3 left in the vault”
- ◆ Stars, thumbs votes and “% recommend”; reviews with the verified-purchase badge
- ◆ Comparable boutique estimate, instalments, certificate of authenticity
- ◆ Manager-added pieces get a page automatically (product.html?id=...)

INFORMATION · DONE

RATINGS · LIVE



# Ratings You Can Trust



## Star ratings

Every piece shows an average out of 5 and the review count. Patron reviews posted through the site are blended into the seeded baseline, so the number moves as real reviews arrive.



## Written reviews

Signed-in patrons write a review (1–5 stars, up to 240 characters) on any piece — `POST /api/reviews`. A patron may delete their own; a manager may delete any.



## Verified purchase

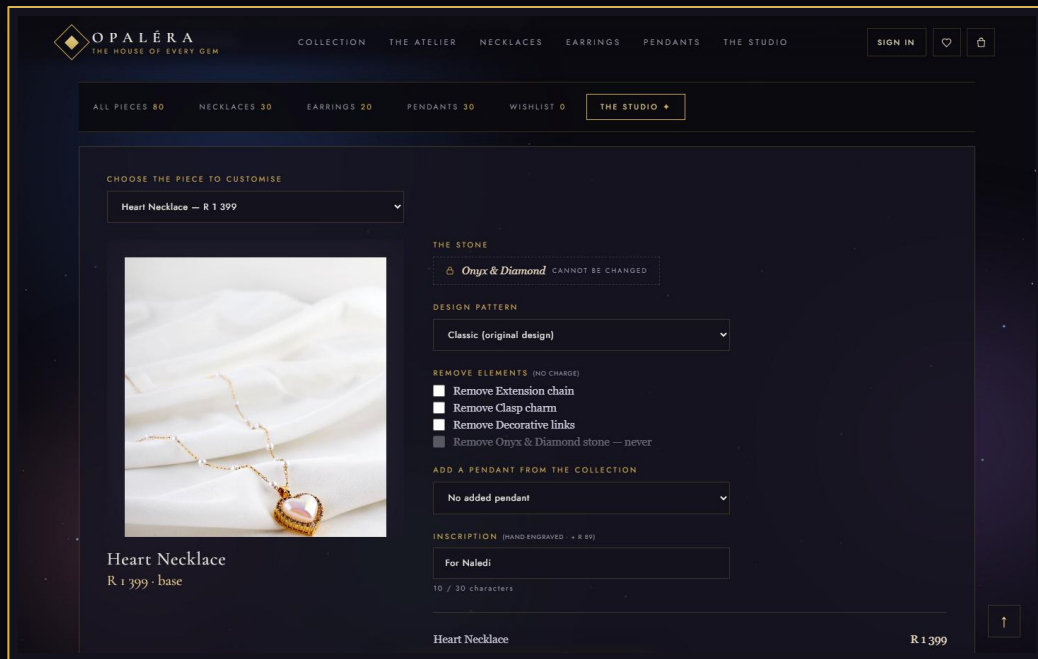
When a review is posted the server looks through that patron's orders; if the piece is there the review is stamped VERIFIED. Nobody can fake the badge from the browser.



## Thumbs & “% recommend”

One vote per browser per piece, tap again to undo. The console's Engagement tab ranks pieces by recommend share and shows the split by collection.

# The Studio — Customisation, With Rules



- ◆ Choose any piece, then: design pattern, removable elements, add a pendant from the collection, a hand-engraved inscription (+ R 89)
- ◆ The stone is never changed — locked on screen; signature pieces allow the inscription only
- ◆ 30-character inscription with a content filter
- ◆ Priced live; the customised piece is carried to the bag, the invoice and the order record

BONUS

# Checkout: Quantities, Rules & VAT

- ◆ Bag with quantities, capped by real stock; customisations itemised; saved per account on the server
- ◆ Promo code (GEM10 / WELCOME10), complimentary gift wrap, delivery address with all nine provinces
- ◆ Card (simulated, never stored), Instant EFT or pay-at-counter — totals recomputed on the server so the client can't tamper
- ◆ Stock is deducted on the server the moment the order is placed

CART · DONE

TRANSACTIONS · DONE

The screenshot displays the 'SECURE CHECKOUT' page for 'OPALÉRA THE HOUSE OF EVERY GEM'. The 'Payment' step is active, with a progress bar showing '1 YOUR BAG', '2 DELIVERY & PAYMENT' (current), and '3 CONFIRMATION & INVOICE'. The page is divided into three main sections: 'Your Order', 'Delivery Address', and 'Payment'.

**Your Order:**

| Item              | Quantity | Price   |
|-------------------|----------|---------|
| Heart Necklace    | 1        | R 1 399 |
| Sapphire Necklace | 2        | R 3 518 |

Below the order items, there is a 'PROMO CODE (TRY: GEM10)' field with an 'APPLY' button. A checkbox for 'Wrap this order as a gift (complimentary)' is checked.

**Summary:**

|                             |                |
|-----------------------------|----------------|
| Items                       | R 4 917        |
| VAT included (15%)          | R 641          |
| Delivery (insured, tracked) | Free           |
| <b>Total due</b>            | <b>R 4 917</b> |

**Delivery Address:**

RECIPIENT FULL NAME: Demo Patron

STREET ADDRESS: 12 Rosebank Avenue

CITY / TOWN: [Empty] POSTAL CODE: [Empty]

PROVINCE: Gauteng PHONE: 071 234 5678

**Payment:** — for Demo Patron

# Eight Rules, Enforced Where It Counts



## Promo codes

GEM10 / WELCOME10 take 10% off the items. The discount is capped at the items total and re-applied server-side.



## 15% VAT inside the price

Prices are VAT-inclusive; VAT = total  $\times$  15 / 115 is shown on the bag, the checkout and the invoice.



## No over-ordering

Server rule: an order for more units than the vault holds is refused — “Only 18 of Sapphire Necklace left”. Retired pieces are refused too.



## Quantity capped by stock

The bag's + button stops at the live stock figure; stock is deducted on the server at checkout.



## Signature pieces

The stone is never customisable; signature pieces allow an inscription only.



## Inscription filter

30 characters, hand-engraved at + R 89, with a content filter on the text.



## Delivery & collection

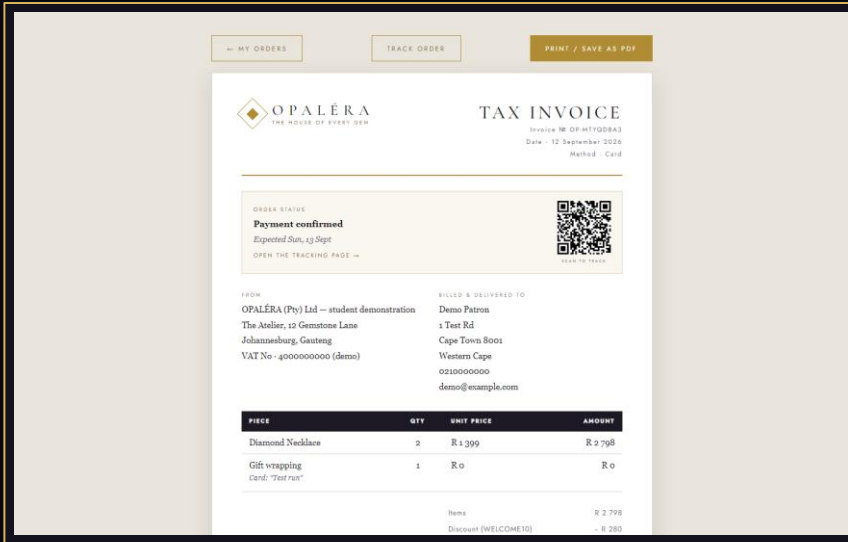
Delivery address requires one of the nine provinces; pay-at-counter orders skip the address and are reserved, not paid.



## Verified reviews

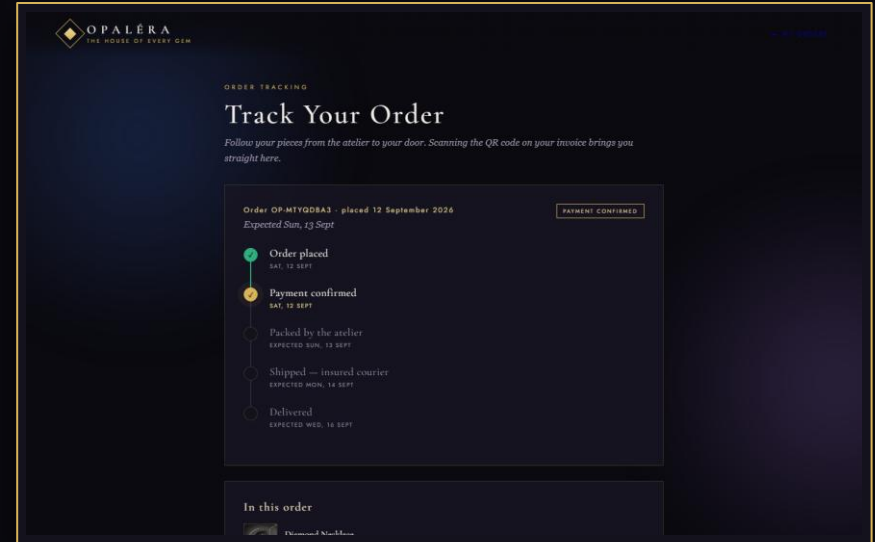
A review is marked verified only if the server finds the piece in that patron's own orders.

# A Tax Invoice — and Where Your Order Is



Tax invoice · generated from the server's order record: letterhead, invoice No, date, method, line items with inscriptions, discount and VAT broken out, print / save as PDF.

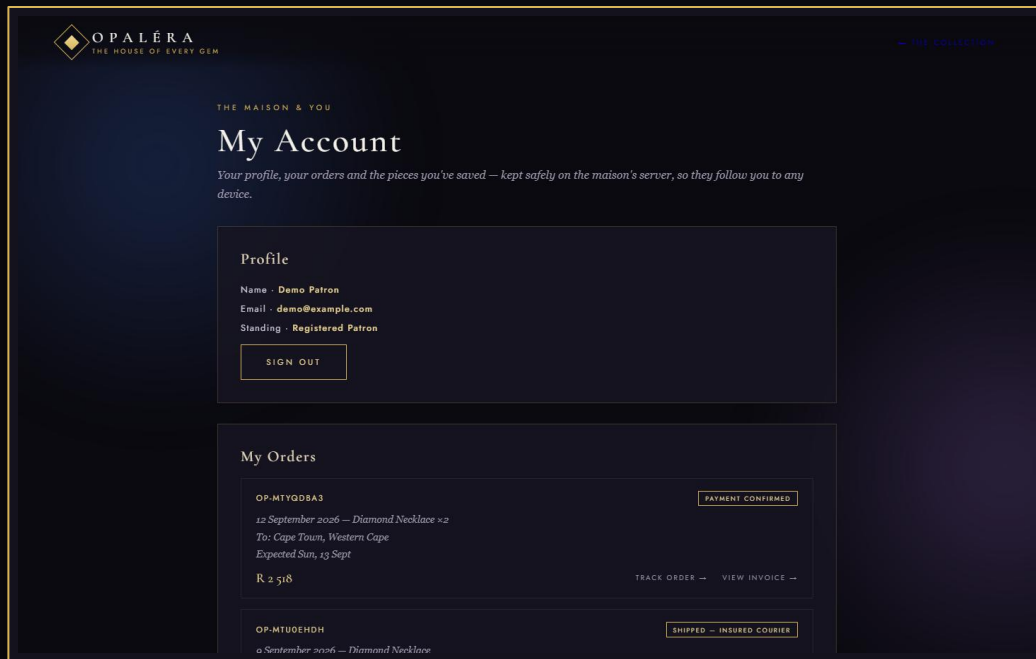
INVOICES · DONE



Tracking timeline · Placed → Paid → Packed → Shipped → Delivered (a collection path for pay-at-counter), advancing with time; the QR on the invoice opens it.

BONUS

# Three Kinds of Visitor, Three Doors



- ◆ Guest — browses, searches, votes, reads reviews
- ◆ Patron — registers (bcrypt-hashed password, httpOnly cookie session); wishlist and bag saved on the server; orders, verified reviews, order history & invoices
- ◆ Manager — a separate console session (role “manager” on the server); patron accounts are refused; managers can also remove a patron account
- ◆ Forgot-password: one-time 6-digit code, single use, 15-minute expiry; show-password eye on every field

REGISTRATION · DONE

LOGIN · DONE

ROLES · DONE

# Sign In, Stay In, Reset Safely



## Passwords are never stored

Each account keeps a random salt and an script hash; sign-in compares hashes in constant time. Minimum 6 characters; emails validated and de-duplicated.



## Cookie sessions

A random 32-byte token in an httpOnly, SameSite cookie — invisible to page scripts — valid 30 days. Log out destroys it on the server.



## Two doors, two cookies

Patron sessions and manager sessions use different cookies, so a customer signed in on the storefront and the manager in the console never collide.



## Forgot password

Request a 6-digit code (email delivery is simulated for the demo), 15-minute expiry, 5 attempts, single use; every other session is signed out on reset.



## Bags that follow you

A guest's bag and wishlist are merged into the account on sign-in and pushed to the server; the pages mirror the signed-in patron locally for instant paint.



## Roles on the server

patron / manager stored per account; manager-only routes answer 401 without a session and 403 to a patron. The backend folder is never served over HTTP.

# One Server, Eight Collections, 24 Endpoints



## users

email · names · script salt+hash · role  
· createdAt



## sessions

token → email · expiry (patron &  
manager scopes)



## products

80 pieces: category · name · price ·  
img · desc · stock · hidden



## orders

ref · items · totals · VAT · promo · gift ·  
address · method · status



## reviews

pid · stars · text · verified · patron



## wishlists · carts

per patron, merged on sign-in



## resets

one-time codes with expiry & attempt  
limit

*Reset the demo: delete db.json — the  
catalogue re-seeds and the manager  
account is recreated on the next run.*

### AUTH

POST signup · login · forgot · reset · logout · GET me · admin/me

### CATALOGUE

GET products · POST / PUT /:id / DELETE /:id (manager)

### PATRON

GET/PUT cart · GET/PUT wishlist · GET/POST orders · GET reviews, reviews/:pid · POST / DELETE reviews

### MANAGER

GET admin/orders · GET admin/users · DELETE admin/users/:email

# A Back Office in the Same World



- ◆ Own manager sign-in — verified by the server; patrons are refused
- ◆ Six KPI cards that count up: pieces sold with an improving / declining verdict, revenue, orders & average order, different products sold, stock on hand, new patrons
- ◆ Charts grow into place: units per month with the revenue line, sales-by-collection donut, units by province, users per day
- ◆ Live indicator, greeting and toasts — the same midnight-and-gold as the storefront; keyboard-navigable tabs

PRODUCT MGMT · DONE

REPORTS · DONE

# Reports: Data Decides

- ◆ Filter by period (all time · 30 / 90 days · this year · custom months), collection, province and source — every figure recalculates
- ◆ Values: pieces sold + growth verdict, revenue, orders & average order, different products sold (text control), stock on hand & vault value by collection, registered users per day, best sellers
- ◆ Sample history is clearly labelled; live orders from the database merge on top
- ◆ Every chart is drawn by hand on a canvas — no chart library



# Add, Edit, Retire, Delete

The screenshot shows the Opaléra console interface. At the top, there's a header with the logo, 'Good afternoon, Ompha', the date 'SUN, 13 SEPTEMBER', and buttons for 'LIVE', 'VIEW STOREFRONT', and 'SIGN OUT'. The main content area is titled 'EDIT PIECE #9 - HEART NECKLACE'. It contains a form with fields for 'NAME' (Heart Necklace), 'COLLECTION' (Necklace), 'PRICE (R, VAT INCL.)' (1399), and 'STOCK' (3). There's also an 'IMAGE URL' field with a placeholder image and a 'DESCRIPTION' field with a text area. Below the form are 'SAVE CHANGES' and 'CANCEL' buttons. At the bottom, there's a section titled 'ALL PIECES' with a search bar and filters for 'All collections' and 'Any status'. Below this is a table listing three pieces: Sapphire Necklace, Cyan Necklace, and Diamond Necklace, each with its own set of 'EDIT', 'RETIRE', and 'DELETE' buttons.

| # | PIECE             | COLLECTION | PRICE   | STOCK | STATUS | ACTIONS            |
|---|-------------------|------------|---------|-------|--------|--------------------|
| 1 | Sapphire Necklace | NECKLACE   | R 1 759 | 16    | LIVE   | EDIT RETIRE DELETE |
| 2 | Cyan Necklace     | NECKLACE   | R 3 199 | 16    | LIVE   | EDIT RETIRE DELETE |
| 3 | Diamond Necklace  | NECKLACE   | R 2 119 | 9     | LIVE   | EDIT RETIRE DELETE |

- ◆ Add a piece: name, collection, price, stock, photo (with preview) and description → POST /api/products; it gets a product page automatically
- ◆ Edit any piece in place (shown: Heart Necklace) → PUT /api/products/:id; the storefront and its product page update immediately
- ◆ Inline stock control; low-stock and out-of-stock flags; retire hides a piece while keeping its history
- ◆ Delete removes it from the database; search and status filters over all 80 pieces

ADD · 3

EDIT · 5

DELETE · 2

# Every Order, Every Patron

Good afternoon, Ompha + SUN, 13 SEPTEMBER LIVE VIEW STOREFRONT SIGN OUT

REPORTS INVENTORY & PIECES **ORDERS** PATRONS ENGAGEMENT

## Orders

Every order in the database, from every patron, newest first.

Search reference or email Any method 11 of 11 orders

| REFERENCE  | DATE       | CUSTOMER         | DESTINATION             | ITEMS | VAT   | TOTAL   | METHOD      | STATUS         |
|------------|------------|------------------|-------------------------|-------|-------|---------|-------------|----------------|
| OPMTZPRLSW | 2026/09/13 | demo@heample.com | Cape Town, Western Cape | 2     | R 412 | R 3 166 | Cash        | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/13 | demo@heample.com | Cape Town, Western Cape | 2     | R 328 | R 2 518 | Cash        | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/13 | demo@heample.com | Cape Town, Western Cape | 2     | R 328 | R 2 518 | Cash        | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/09 | demo@heample.com | Pretoria, Gauteng       | 1     | R 287 | R 2 199 | Instant EFT | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/09 | demo@heample.com | Johannesburg, Gauteng   | 1     | R 136 | R 1 039 | Cash        | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/05 | demo@heample.com | Johurg, Gauteng         | 2     | R 439 | R 3 518 | Cash        | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/05 | demo@heample.com | Isidore collection      | 1     | R 0   | R 1 759 | Cash        | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/03 | demo@heample.com | Johurg, Gauteng         | 1     | R 182 | R 1 299 | Instant EFT | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/03 | demo@heample.com | Johurg, Gauteng         | 2     | R 793 | R 6 078 | Instant EFT | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/02 | demo@heample.com | Johannesburg, Gauteng   | 1     | R 229 | R 1 759 | Instant EFT | PAID Invoice ✓ |
| OPMTZPRLSW | 2026/09/02 | demo@heample.com | Johannesburg, Gauteng   | 3     | R 601 | R 4 605 | Instant EFT | PAID Invoice ✓ |

Orders · every order from every patron, newest first: reference, date, customer, destination, items, VAT, total, method, status; search by reference or email, filter by method; open the invoice.

REPORTS · DONE

Good afternoon, Ompha + SUN, 13 SEPTEMBER LIVE VIEW STOREFRONT SIGN OUT

REPORTS INVENTORY & PIECES ORDERS **PATRONS** ENGAGEMENT

## Patrons

Registered accounts and when they joined; the "Registered users per day" chart on the Reports tab follows the same period filter.

2 registered - 1 manager

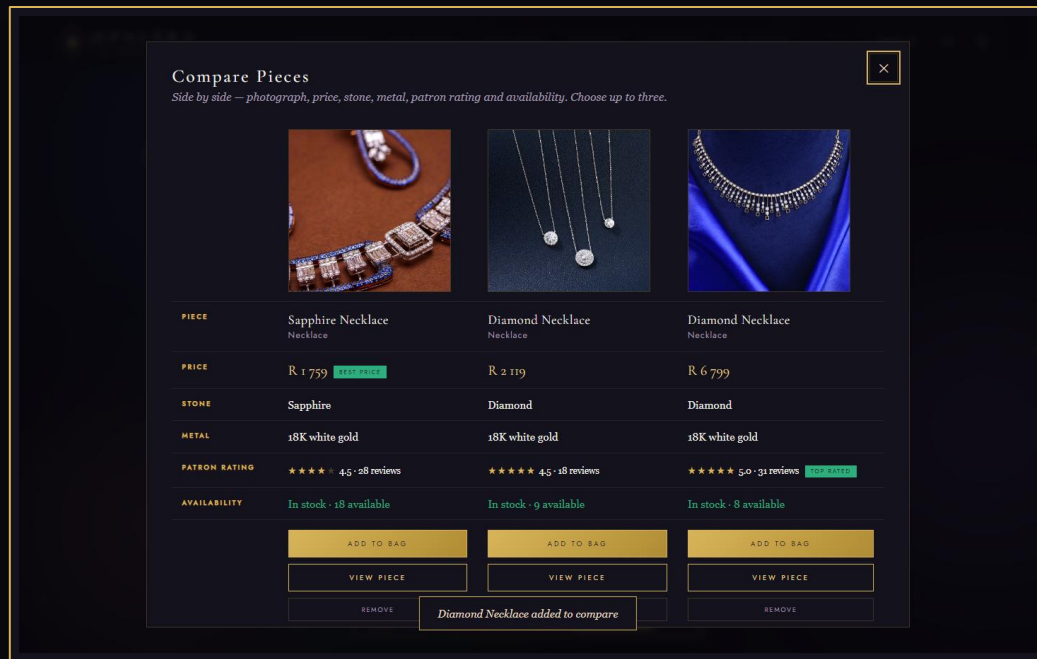
| NAME          | EMAIL                 | ROLE    | JOINED       | ORDERS | SIGN-UPS BY DAY     |
|---------------|-----------------------|---------|--------------|--------|---------------------|
| Ompha Manager | manager@opalera.co.za | MANAGER | 09 Sept 2026 | 0      | Wed, 09 Sept 2026 1 |
| Demo Patron   | demo@heample.com      | PATRON  | 02 Sept 2026 | 11     | Wed, 02 Sept 2026 1 |

Patrons · every account with role, sign-up date and order count; sign-ups by day beside it. Managers can remove a patron account (managers are protected).

USER MGMT · DONE

# Compare, Track, Scan

- ◆ Compare pieces — tick up to three; a side-by-side table of photo, price (“Best price”), stone, metal, rating (“Top rated”) and availability, with add-to-bag in place
- ◆ Order tracking — a live timeline per order, with status chips and ETA on the account page
- ◆ QR code on every invoice — printed invoices open the order's tracking page from a phone
- ◆ Also: the Studio customiser, and a motion system that respects reduced-motion



# Built to Be Trusted



## 64 automated checks — local and live

Site & static blocking, sign-up / login / reset, cart, wishlist, orders with server-side totals, the stock rule, reviews, roles, product CRUD and validation, then cleanup of its own test data — all passing against localhost and against opalera.onrender.com.



## Browser-verified pages

Every page loaded with zero console errors; compare, tracking, invoice QR, account and the console exercised end to end; keyboard navigation checked.



## Security by default

script password hashes with per-user salt, constant-time comparison, httpOnly SameSite cookies, server-recomputed totals and stock rules, the backend folder never served over HTTP, role checks on the server, a curated seed database with no tokens.



## Accessible & considerate

Reduced-motion switch, WAI-ARIA tabs with arrow keys, live regions, focus rings, colour contrast checked; a multi-agent review found and fixed 29 defects before release.

# Live on Render, Runs Anywhere Node Does

## RUN IT LOCALLY

```
node opalera-backend/server.js
```

- ◆ Then open <http://localhost:3000> — it redirects to the storefront
- ◆ One server hosts the whole site and its API; no installs, no build step
- ◆ Data lives in `opalera-backend/data/db.json` (or `DATA_DIR`), created on first run

## DEMO ACCOUNTS

Patron [demo@example.com](mailto:demo@example.com) / `demo123`

Manager

[manager@opalera.co.za](mailto:manager@opalera.co.za) / `Opalera2026`



## Live site

[opalera.onrender.com](https://opalera.onrender.com)

Storefront [/pages/jewellery\\_Html.html](/pages/jewellery_Html.html) · Console </pages/admin.html>



## Source & workflow

[github.com/Thrivi17/Opalera](https://github.com/Thrivi17/Opalera)

Both members commit to main as collaborators; every push redeploys the live site within minutes. The seed database, docs and presentation live in the repo.

ALL PIECES 80

NECKLACES 30

EARRINGS 20

PENDANTS 30

WISHLIST 0

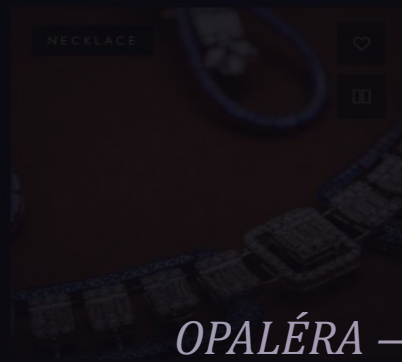
THE STUDIO +

Search the archive

Every Price

Curated Order

SHOWING 80 OF 80 PIECES



Sapphire Necklace

★★★★ 4.4 25

R 1 759

VIEW PIECE —



Cyan Necklace

R 3 199

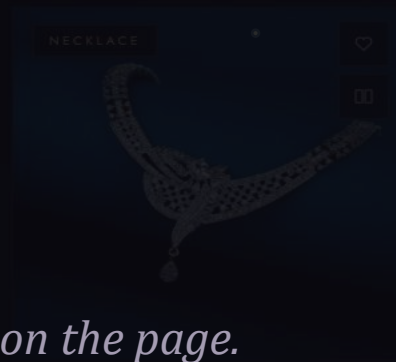
VIEW PIECE —



Diamond Necklace

R 2 119

VIEW PIECE —



Diamond Necklace

R 3 199

VIEW PIECE —

Tsongorerera RJN · Thedzisani KOP · Madhavan T · Mtshare B

